
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead when applied to long videos, primarily because video inputs are represented by large sets of visual tokens. Existing efficiency methods typically reduce this cost by pruning or merging tokens according to estimated importance or feature similarity. However, such methods largely omit a complementary property of video data: temporally localized changes and turning points. They also do not provide a unified mechanism for jointly modeling spatial and temporal token relations. To address these limitations, we formulate a technical perspective in which similarity identifies redundant content, whereas difference identifies event-critical transitions. On this basis, we introduce ST-SimDiff, a training-free token selection framework. The method first constructs a spatio-temporal graph over visual tokens to represent their local spatial and temporal dependencies in a common structure. It then applies a parallel dual-selection procedure: 1) similarity-based selection performs community detection and retains representative tokens to compress persistent or static content; 2) temporal difference-based selection detects content-transition points and preserves tokens associated with salient dynamic changes. The resulting subset preserves both stable visual context and dynamic event information under a reduced token budget. Extensive experiments show that ST-SimDiff outperforms state-of-the-art compression approaches while substantially decreasing computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The development of Large Language Models (LLMs) has accelerated progress in Large Vision-Language Models (LVLMs), which now provide strong capabilities for image and video understanding (Zhang et al., 2024c; Liu et al., 2024c). In video processing pipelines, current LVLMs usually sample an input video into a frame sequence and encode each frame into hundreds or thousands of visual tokens before passing them to the LLM (Bai et al., 2025; Li et al., 2024). Although this representation is effective, the token count increases rapidly as video duration and spatial resolution grow. Consequently, inference requires substantial computation and memory, which limits the deployment of LVLMs in long-video analysis, real-time interaction, and other resource-constrained scenarios (Fu et al., 2024b).

Existing work addresses this bottleneck through several classes of LVLM efficiency methods. The first class performs importance-based pruning. FastV (Chen et al., 2024) observes redundancy in cross-layer attention distributions during visual processing and removes visual tokens with low attention scores in deeper model layers to reduce computation. The second class performs similarity-based merging or selection. These methods exploit high feature similarity among visual tokens, either across adjacent video frames or within individual frames. FrameFusion (Fu et al., 2024b) merges similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) selects dominant tokens directly at the vision-encoder stage.

Despite their effectiveness for visual token reduction, existing methods have two technical limitations. First, many token compression strategies model either spatial correlations within a frame or temporal correlations between corresponding positions across frames, rather than jointly modeling

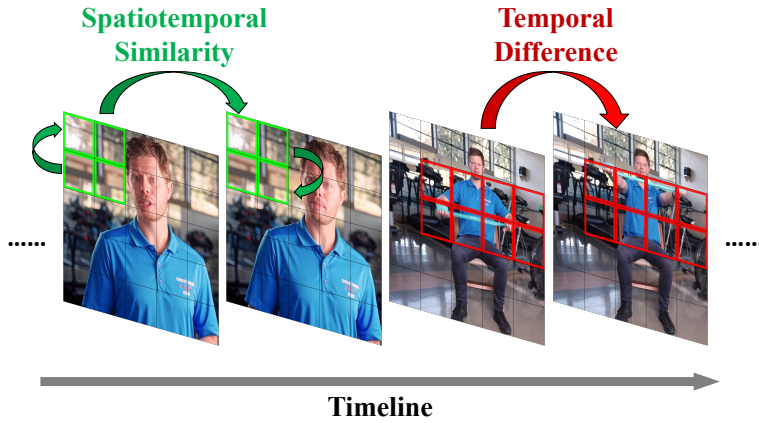


Figure 1: Core motivation of ST-SimDiff. Efficient video understanding requires the simultaneous treatment of two token-level structures: spatiotemporal similarity (left), which supports compression of redundant information within frames and across adjacent frames, and temporal difference (right), which identifies actions or events that determine the video narrative.

spatio-temporal dependencies. This separation restricts their ability to represent dynamic events whose evidence is distributed over both space and time. Second, these methods primarily optimize for informational commonality, such as token importance or feature similarity, and therefore under-specify changes and differences. Since video semantics are often determined by turning events, a compression rule based only on similarity can suppress abrupt actions and thereby distort the downstream interpretation.

Motivated by this analysis, we propose the following principle: similarity should be used to identify redundancy, while difference should be used to identify key moments. An effective video token compression algorithm should therefore satisfy two requirements simultaneously: it should represent stable video content with a small number of tokens and preserve tokens associated with salient changes. To this end, we introduce **ST-SimDiff**, a dual token selection framework defined on a spatio-temporal graph. The first component is **similarity-based representative token selection**: all visual tokens in a video are treated as graph nodes, and edges are constructed according to spatio-temporal feature similarity. A graph community detection procedure identifies strongly connected token clusters corresponding to stable and persistent visual elements, and central tokens within each cluster are retained as representatives. The second component is **difference-based event token selection**: temporal edges in the graph are examined to detect sharp decreases in similarity between corresponding tokens from adjacent frames. When such a decrease occurs, the associated token is treated as a critical event token and retained. The final token set is obtained by merging the representative and event-token sets before input to the LLM. Our contributions can be summarized as follows:

- We introduce a token compression perspective in which token similarity and token difference are treated as equally important signals for efficient VLM inference.
- We design a spatio-temporal graph framework that models video-token relations in a unified structure and supports a dual selection strategy for representative tokens and pivotal event tokens.
- We conduct extensive experiments on video understanding benchmarks, demonstrating that ST-SimDiff substantially reduces the visual token budget while maintaining or improving task performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid development of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has driven substantial progress in multimodal understanding and has led to a broad family of Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These systems typically encode images or video frames into visual tokens with a pre-trained vision encoder and then provide those tokens to an LLM. Alignment modules such as MLP projectors (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023) are commonly used to connect the vision encoder and language model. This architecture enables strong performance across multi-modal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a) and is particularly relevant for video understanding, where recent LVLMs are designed to process longer, higher-resolution, and more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

The same representational capacity also introduces significant computational pressure in video settings. Because a video must be represented as a temporally ordered sequence of frames, the number of visual tokens can far exceed the token count used for static images. This growth degrades both training and inference efficiency. Efficient processing of large video-token sequences, and the corresponding reduction of computational overhead, therefore remains a central technical problem in LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Given the computational cost of video-token processing, visual token compression has become an important mechanism for improving LVLM efficiency. Existing methods generally define compression criteria using either token importance or token similarity.

Early approaches primarily perform importance-based pruning, in which visual tokens estimated to contribute less to model outputs are removed. These methods commonly use attention scores or related feature statistics to quantify token importance, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Later work (Shang et al., 2024; Ye et al., 2025) incorporates token similarity to reduce redundancy while retaining informative content. This line of work recognizes that visually important tokens may still be mutually redundant. For example, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning. VisionZip (Yang et al., 2025) selects dominant tokens and merges contextual information, and VisPruner (Zhang et al., 2025b) uses visual cues to remove redundant elements.

Although visual-token redundancy has been studied extensively, existing methods seldom model global spatio-temporal similarity relations among tokens. In video data, redundancy occurs both spatially within a frame and temporally across frames. Similarity-driven methods such as FrameFusion mainly emphasize temporal similarity between adjacent frames and do not fully exploit the richer spatio-temporal dependency structure of video tokens. This omission limits their capacity to use the inherent redundancy of video representations, particularly under aggressive token-reduction budgets.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video as a complete sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This representation is computationally expensive because the self-attention complexity scales quadratically with the token count N , i.e., $O(N^2)$. The objective is therefore to define an efficient token selection function $f(\cdot)$ that maps the full token sequence T_{full} and a compression ratio r to a reduced subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The selected subset should maximize downstream LVLM performance subject to the budget constraint $|T_{\text{sub}}| \approx r \cdot N$. The central design problem is to specify $f(\cdot)$ such that T_{sub} retains the information most relevant to the downstream video-understanding task.

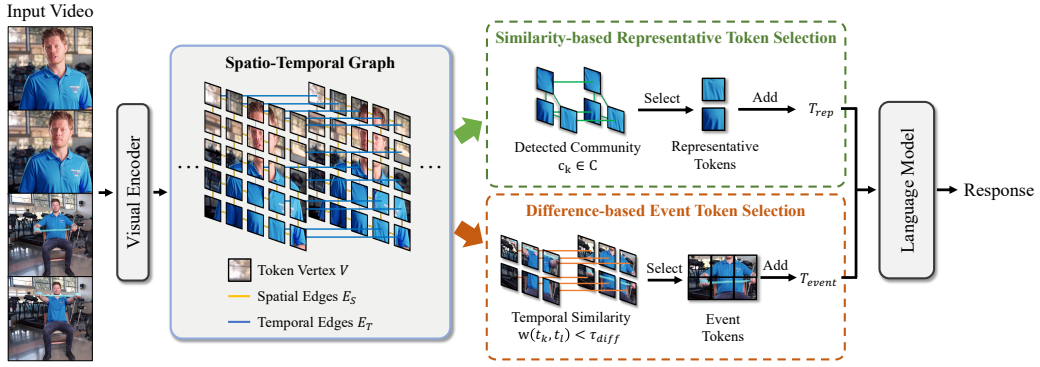


Figure 2: Overview of ST-SimDiff. The framework comprises three technical stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As shown in Figure 2, ST-SimDiff consists of two parallel components applied to a spatio-temporal graph constructed from the visual tokens of a video: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The SRTS module compresses stable and redundant video content. It applies community detection to the graph so that highly similar tokens are grouped into clusters corresponding to persistent visual elements, such as static backgrounds. Within each cluster, tokens with high centrality are retained to form the representative set T_{rep} . The DETS module preserves dynamic transitions. It analyzes temporal graph edges and identifies positions at which the similarity between corresponding tokens in adjacent frames decreases sharply. These high-difference locations are treated as turning points, and the tokens associated with the transitions are retained as event tokens, forming the set T_{trans} . By combining these two selection outputs, ST-SimDiff constructs the final subset $T_{sub} = T_{rep} \cup T_{trans}$, which is compact while retaining both persistent visual context and event-specific temporal evidence.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder maps it to a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ is associated with a spatio-temporal location and a feature vector $\mathbf{x}_k$. The location of t_k is specified by its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$.

We represent token relations with a spatio-temporal graph $G = (V, E)$, where the vertex set V corresponds to all tokens in T . The edge set E is decomposed as $E = E_S \cup E_T$, where E_S encodes spatial adjacency and E_T encodes temporal adjacency. The spatial edge set E_S connects tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edge set E_T connects tokens located at the same spatial position in adjacent frames, thereby encoding local temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

For any edge in E , the edge weight $w(v_i, v_j)$ is defined as the cosine similarity between the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph representation encodes local spatial structure and temporal continuity within a unified relational object, which serves as the substrate for the dual-path token selection procedure.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video representations contain substantial spatio-temporal redundancy. In a static scene, background tokens remain highly similar across consecutive frames. For a moving object, the corresponding tokens may occupy different spatial positions but still preserve strong semantic correlation. Existing methods often process temporal and spatial similarity separately, whereas the proposed spatio-temporal graph is designed to represent their joint structure. In this graph, semantically related tokens form dense “communities” or “clusters” according to feature similarity, regardless of their exact frame index or spatial coordinate. Selecting representative tokens from each community therefore provides a direct mechanism for compressing redundant content while retaining semantic coverage.

To identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only the edges with weights above this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its tokens according to centrality. The centrality score $S_c(t_a)$ for a token $t_a \in c_k$ is defined as the average similarity between t_a and all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We then set the intra-community retention rate equal to the global compression ratio r and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the largest centrality scores in each community c_k . This filtering rule applies uniformly to all communities, including single-node communities that correspond to unique tokens; in such cases, the sole node is retained. Formally, the representative token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(|c_k| \cdot r) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. This procedure compresses static and persistent video content by retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity characterizes the regular structure of a video, then difference characterizes its event boundaries. A compression method that uses only similarity can represent “what is present” but may fail to represent “what changes.” Video narratives are often determined by changes such as the appearance of a new object, the initiation of an action, or a scene transition. In the proposed graph, these changes correspond to large feature discrepancies between temporally adjacent tokens. Detecting such turning points is therefore necessary for preserving dynamic content and for answering temporally grounded questions such as “when” and “why”.

Temporal difference is especially informative because key events frequently alter visual features at corresponding positions in adjacent frames. Examples include object entry, scene changes, and action starts or ends. We therefore analyze the temporal edges (E_T) in the spatio-temporal graph. For a temporal edge $(v_k, v_l) \in E_T$ that connects adjacent-frame tokens t_k and t_l , we define a dynamic threshold τ (e.g., the 95th percentile of all temporal-edge difference scores). If the difference score for (v_k, v_l) exceeds this threshold, equivalently if $w(t_k, t_l) < \tau_{\text{diff}}$, the subsequent token t_l is treated as a critical event token and retained. Formally, the event-token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This selection rule preserves tokens associated with substantial temporal change. By retaining these transition tokens, the compression process maintains dynamic evidence that could otherwise be removed by redundancy-oriented selection.

4.5 OVERALL REDUCTION PROCESS

The complete reduction procedure first computes the representative token set T_{rep} and the event token set T_{event} in parallel. Their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, is used as the initial candidate set. To satisfy the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we apply a final pruning step. If $|T_{\text{candidate}}|$ exceeds N_{target} , the method removes the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following prior work (Chen et al., 2024), token importance is determined by the attention score computed in shallow LLM layers. This final step enforces the computational budget while retaining the graph-selected information as much as possible.

Overall, the framework combines graph-based token selection with attention-based dynamic pruning. The graph stage uses intrinsic video structure, expressed through similarity and difference, to identify tokens with representative or event-specific value. The attention-pruning stage provides a budget-matching mechanism that adjusts the candidate set to the requested token count. This design yields a compression procedure that is both structurally grounded and adaptive to a specified computational budget.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff can be decomposed into three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The first stage constructs the graph. For each of the N visual tokens, the method computes cosine similarity with spatially and temporally adjacent neighbors. Because each token has a bounded constant number of neighbors, this stage requires one pass over the token sequence and has complexity $O(Nd)$, where d denotes the token feature dimension.

After graph construction, SRTS identifies token clusters. As specified in the implementation (Section 5.2), we use a connected components algorithm, which has near-linear complexity $O(N + |E|)$. Since the number of edges $|E|$ is at most $3N$, this term reduces to $O(N)$. The subsequent within-community filtering step could be expensive if implemented by all-pairs similarity computation inside each community c_k , which would require $O(|C| \cdot |c_k|^2 \cdot d)$ operations. To avoid this cost, we impose a maximum community-size constraint. If a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), it is partitioned before filtering. This constraint prevents the filtering stage from dominating the total complexity. Although standard graph clustering algorithms such as Louvain and Leiden provide more expressive community definitions, their computational costs are higher and approach $O(N \log N)$. We therefore use the simpler connected components procedure to minimize processing overhead.

The DETS module performs a single pass over all temporal edges and compares token similarities with the difference threshold to identify significant temporal changes. This stage has linear complexity $O(Nd)$.

Consequently, the total complexity of ST-SimDiff is dominated by linear-time operations and is $O(Nd)$. This is substantially lower than the $O(N^2d)$ complexity of the self-attention mechanism in the Large Language Model. The proposed method therefore reduces the token sequence with a small computational footprint relative to the main LVLM inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate the effectiveness of the proposed method, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare ST-SimDiff with state-of-the-art video-token compression techniques. The baseline set covers multiple compression paradigms, including **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For calibration, we

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video-understanding benchmarks using LLaVA-Video-7B with 64 input frames (%). The best result among the compared methods is marked in **bold**.

also report the original uncompressed model (Vanilla) as the performance upper bound and Random Sampling with the same token count as a lower-bound reference.

Benchmarks To assess long-video understanding, we use three challenging benchmark datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME provides a comprehensive evaluation protocol for foundational video understanding and contains diverse video types divided into Short, Medium, and Long duration categories. LongVideoBench evaluates retrieval and reasoning over long videos through a “referring reasoning” task, with video lengths of 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering benchmark for long-term first-person videos and measures the ability to infer character intentions and causal action chains.

5.2 IMPLEMENTATION DETAILS

All experiments are executed on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04. Inference evaluation is implemented with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow the original configuration with 64 input frames. To control for computational efficiency and resource usage, we also set the NVILA input frame count to 64. The similarity threshold is fixed to $\tau_{\text{sim}} = 0.8$, and the difference threshold is fixed to $\tau_{\text{diff}} = 0.2$. We evaluate token compression rates of $r = 30\%$ and $r = 50\%$. In the practical implementation, we use connected component finding as the community detection algorithm to balance runtime cost and performance gain.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We compare ST-SimDiff with representative efficient video language model (VLM) compression methods. The evaluation covers three long-form video-understanding benchmarks, VideoMME, LongVideoBench, and EgoSchema. To examine generalization across model backbones, we report

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video-understanding benchmarks using NVILA-Video-8B with 64 input frames (%). The best result among the compared methods is marked in **bold**.

results for LLaVA-Video-7B and NVILA-8B under token retain ratios of 30% and 50%. The corresponding results are reported in Table 1 and Table 2.

(1) ST-SimDiff consistently achieves the strongest performance across the tested configurations. At a 50% token retain ratio, the overall performance on both base models exceeds that of competing compression algorithms and, on some benchmarks, matches or surpasses the original model using 100% of the tokens. This result indicates that the selected token subset can preserve task-relevant evidence more effectively than existing reduction criteria.

(2) Among importance-based baselines such as FastV and FasterVLM, FasterVLM generally provides stronger results. However, importance-only methods do not explicitly remove temporal redundancy and can retain tokens that are individually important but repetitive across time, which limits compression efficiency. Hybrid baselines such as PruMerge, VisionZip, and FrameFusion are more competitive, with FrameFusion usually obtaining the best overall baseline performance. Nevertheless, these methods still optimize mainly for representative “commonality” information and may under-preserve narrative evidence associated with “turning points” and “abrupt events”.

(3) Overall, the baseline methods reflect progressively stronger modeling of video “redundancy,” but two limitations remain. First, they lack a joint analysis of spatio-temporal similarity among visual tokens, which makes complex redundancy patterns difficult to capture. Second, they do not explicitly perform “difference” detection and therefore do not actively preserve changes that define plot progression. ST-SimDiff improves performance by constructing a spatio-temporal graph for unified relation modeling and by adding a dedicated difference-based event selection mechanism.

5.4 ABLATION STUDY

To quantify the contribution of each component in ST-SimDiff, we conduct a set of ablation studies. The Baseline uses only fundamental importance-based pruning. We then add the proposed similarity selection module (+Sim) and difference selection module (++Diff) in sequence to isolate their

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation results for the proposed token selection modules (%).

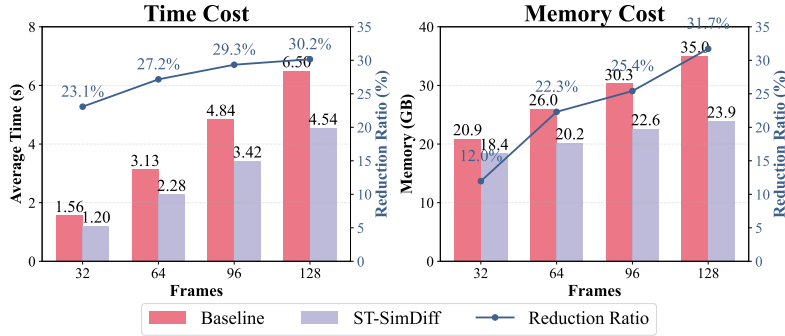


Figure 3: Computational cost comparison between ST-SimDiff and the baseline LLaVA-Video model, measured by average inference time and peak GPU memory usage under a 30% token budget.

individual effects. All ablations are evaluated at token retain ratios of 30% and 50%, and the results are shown in Table 5.

(1) We first evaluate the +Sim module. Adding similarity modeling to the baseline consistently improves performance. We further decompose the similarity strategy into spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) variants. The results show that joint modeling of spatio-temporal token associations provides the most effective redundancy compression.

(2) After selecting the joint spatio-temporal similarity configuration, we add the difference selection module (++Diff) to obtain the full framework. This module yields a substantial performance increase, especially at the more aggressive 30% retain ratio. The result indicates its complementary function: +Sim removes redundant structure, whereas ++Diff identifies and preserves unique event tokens that define the video narrative. At the 50% retain ratio, the marginal gain is smaller because the less aggressive +Sim configuration is more likely to retain these event tokens already, and overall performance is closer to the upper bound. This analysis supports the central design principle that effective compression requires both similarity-based redundancy reduction and difference-based event preservation, particularly under stringent token budgets.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify practical efficiency, we measure average inference time and peak GPU memory usage for ST-SimDiff under a 30% token budget. The reference system is the uncompressed baseline model (Baseline LLaVA-Video). Figure 3 reports results for video inputs ranging from 32 to 128 frames.

Inference Time ST-SimDiff reduces inference latency relative to the uncompressed baseline. As the number of input frames increases, the baseline average inference time rises from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames. Under the same 128-frame setting, ST-SimDiff requires 4.54 seconds on average. The corresponding time-saving rate increases with video length, from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff also lowers peak GPU memory consumption. The baseline memory usage increases with video length, from 20.9 GB at 32 frames to 35.0 GB at 128 frames. By reducing the number of visual tokens passed to the large language model, ST-SimDiff decreases the memory requirement to 23.9 GB for 128-frame videos, corresponding to a 31.7% reduction relative to the baseline. This reduction enables longer-video processing on hardware with more limited memory capacity.

In summary, ST-SimDiff improves both task performance and computational efficiency, reducing inference time and peak memory usage while maintaining strong video-understanding accuracy.

6 CONCLUSION

This paper addresses the computational inefficiency of Multimodal Large Language Models for long-video processing, which arises from large visual-token sequences and from compression strategies that may omit critical content changes. We propose ST-SimDiff, a training-free framework that constructs a spatio-temporal graph to model token associations. The method applies a dual selection strategy: similarity-based selection identifies redundant structure through community detection, and temporal-difference selection preserves tokens at key transition points. Extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff reduces inference time and memory usage while consistently exceeding state-of-the-art compression performance. These results indicate that balancing content representativeness with transition preservation is an effective principle for efficient video representation.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

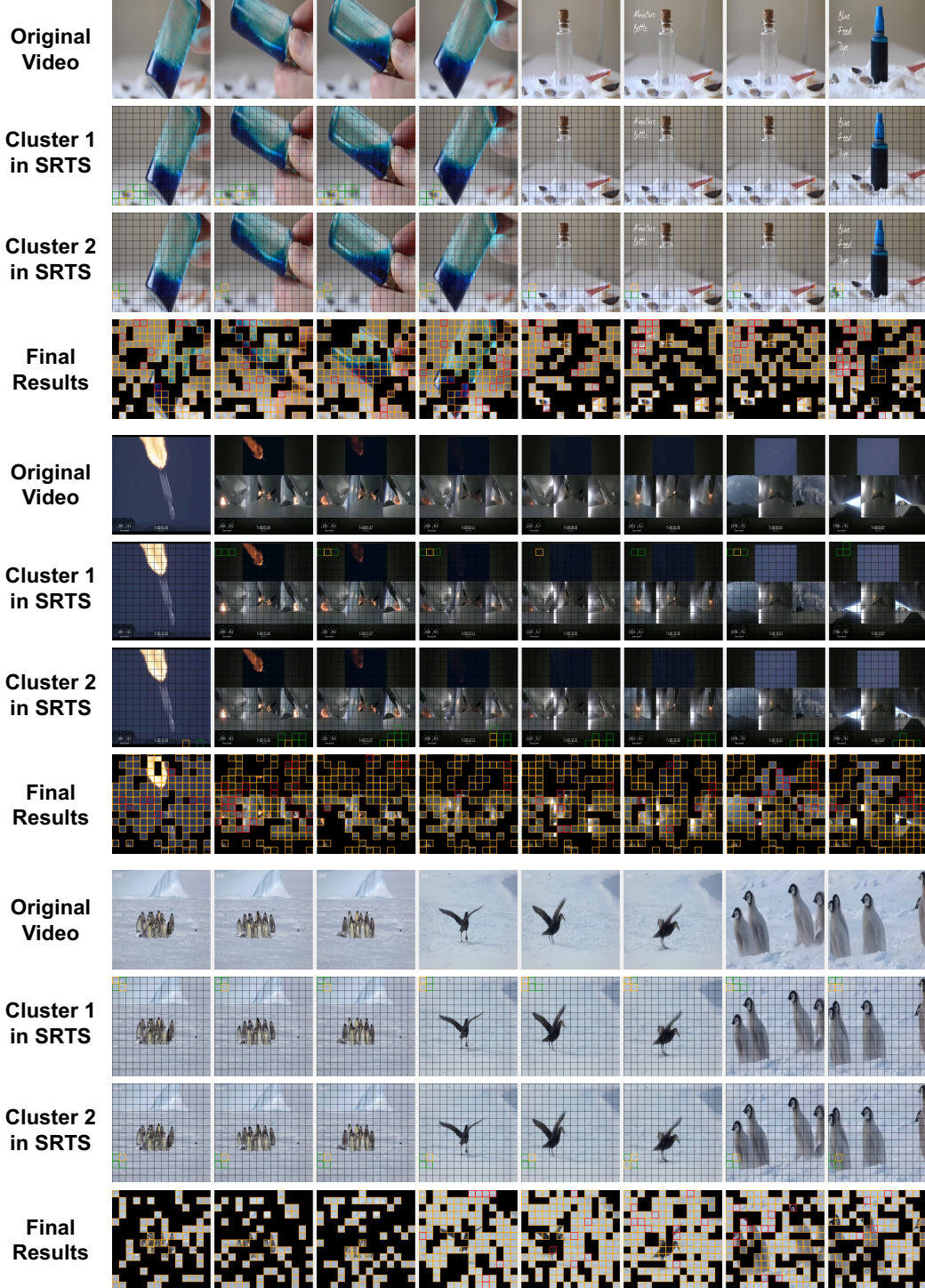


Figure 4: Visualization of the ST-SimDiff token selection process. The figure decomposes selection into communities detected by SRTS, shown as grids in Cluster 1 and Cluster 2, and event-driven tokens detected by DETS. The final panel illustrates the complementary roles of sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

Figure 4 visualizes intermediate and final token selection outputs produced by ST-SimDiff on a video sequence involving dynamic object manipulation. For Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show the behavior of graph-based community detection. Spatially and temporally redundant regions, including the static background and stationary objects, are assigned to coherent communities. The sparse yellow boxes in the final results correspond to central representative tokens selected from these communities, indicating that SRTS compresses stable visual elements while retaining their representative structure.

Difference-based Event Token Selection (DETS) provides the complementary mechanism for dynamic intervals. In the “Final Results” row, tokens selected because of high temporal difference are marked by dense red bounding boxes. These boxes align with the moving hand and changing blue liquid, showing that the method localizes “turning points” at which the temporal difference exceeds the threshold (τ_{diff}). This selection preserves fine-grained action details and rapid visual changes without retaining all static tokens.

The final overlay illustrates the interaction between the two modules. Sparse yellow tokens encode the stable background, whereas dense red tokens encode rapid motion. This separation provides qualitative evidence for the proposed design principle: ST-SimDiff mitigates the blind spots of similarity-only compression by retaining tokens during abrupt changes, while still removing spatio-temporal redundancies that importance-based methods may preserve. The framework therefore balances computational efficiency with preservation of critical visual evidence.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video-understanding benchmarks using Qwen2.5-VL-7B with 64 and 128 input frames (%).

To further evaluate generalization, we conduct additional experiments with the Qwen2.5-VL model. These evaluations use the VideoMME and LongVideoBench benchmarks and test whether the effectiveness of ST-SimDiff transfers beyond the base architectures used in the main experiments.

The results in Table 4 show that ST-SimDiff remains effective on Qwen2.5-VL. With a 30% token retention ratio, the method obtains 62.4 on VideoMME and 56.8 on LongVideoBench. When the retain ratio is increased to 50%, performance increases to 62.9 on VideoMME and 58.4 on LongVideoBench. At the 50% setting, the VideoMME score matches the full-token upper bound, and the LongVideoBench score remains close to the corresponding full-performance result. These observations indicate that the proposed selection rule generalizes across model architectures.

C ABLATION STUDY RESULTS ON NVILA

To assess whether each component generalizes across base models, we conduct additional ablations on NVILA-Video-8B (Liu et al., 2024c). The experimental protocol follows the main-paper ablation: the Baseline contains only fundamental importance-based pruning, after which we add the similarity selection module (+ Sim) and the difference selection module (++ Diff) sequentially.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Component ablation results for ST-SimDiff on the NVILA-Video-8B model (%).

As shown in Table 5, the NVILA results support the same component-level trends observed with LLaVA-Video-7B. Adding the similarity module (+ Sim) to the baseline improves performance at both 30% and 50% token retain ratios. Within this module, we compare spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) similarity. The joint spatio-temporal variant is the strongest configuration; for example, at $r = 50\%$, it achieves 61.4% on VideoMME and exceeds both single-axis variants. This result indicates that redundancy is more effectively identified when spatial and temporal associations are modeled jointly. After selecting the joint spatio-temporal similarity module, we add the dissimilarity selection module (++ Diff) to obtain the complete ST-SimDiff framework. This addition produces the largest gains in the table, confirming that similarity selection and dissimilarity selection provide complementary signals. The NVILA ablation therefore reinforces the central claim that efficient video-token compression should jointly handle redundancy and change.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

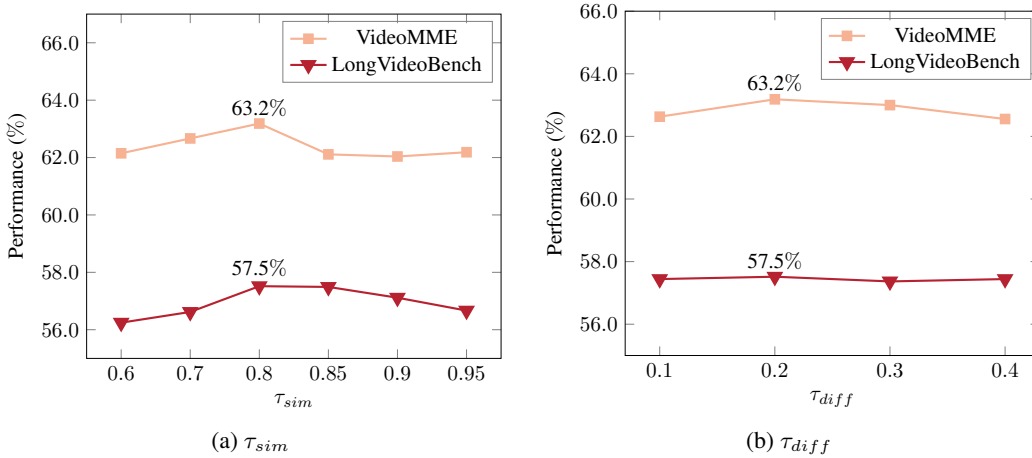


Figure 5: Hyperparameter ablation results for τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

In ST-SimDiff, the similarity threshold τ_{sim} controls representative-token selection, whereas the difference threshold τ_{diff} controls transition-token selection. We analyze their effects through the ablations reported in Figure 6. Performance exhibits a similar pattern for both hyperparameters: it

increases to an optimum and then decreases, while remaining stable over a local range. For τ_{sim} , an overly low value yields imprecise community detection, whereas an overly high value can fragment semantic clusters. For τ_{diff} , an overly low value introduces noisy event detections due to excessive sensitivity, whereas an overly high value can miss salient events. The best configuration is $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, which is therefore used in the remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

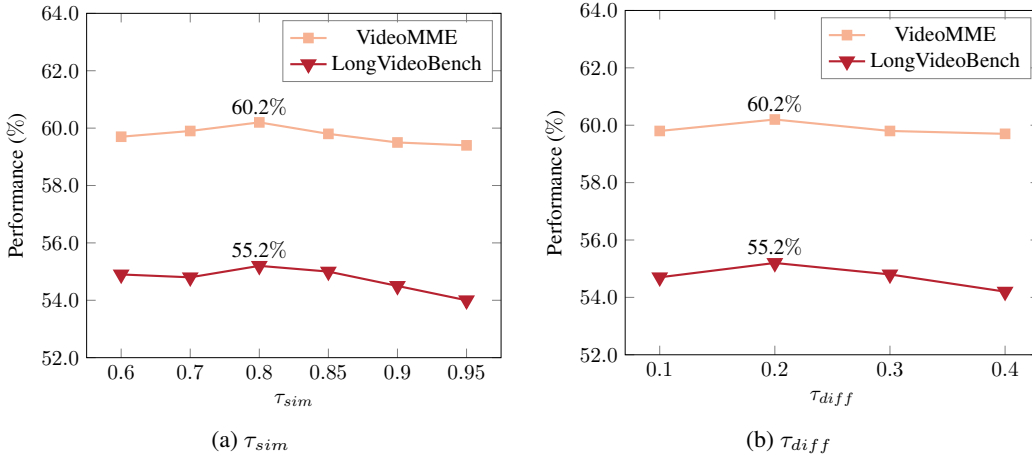


Figure 6: Hyperparameter ablation results for τ_{diff} and τ_{sim} on NVILA-Video-8B.

To evaluate hyperparameter robustness, we conduct additional ablations on NVILA-Video-8B. As shown in Fig. 6, the analysis measures how the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , affect performance on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We evaluate τ_{sim} over the range $[0.6, 0.95]$. Performance is relatively insensitive to this threshold. On VideoMME, the peak score is 60.2% at $\tau_{sim} = 0.8$. On LongVideoBench, the optimum is also obtained at $\tau_{sim} = 0.8$, with a score of 55.2%. Although performance decreases slightly when the threshold is too low or too high, the total variation is small, indicating stability across a broad range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . We evaluate τ_{diff} over the range $[0.1, 0.4]$. As with τ_{sim} , performance remains robust to threshold variation. The best results occur near $\tau_{diff} = 0.2$ on both benchmarks, with 60.2% on VideoMME and 55.2% on LongVideoBench. This result indicates that a difference threshold near 0.2 provides an effective operating point for event-token selection under compression.

In summary, the hyperparameter ablations show that ST-SimDiff is robust to moderate variation in its key thresholds. The results support the use of $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the main experiments, since this configuration consistently provides the best or near-best performance across benchmarks and base models.